

Benchmarking Lexical, Distributional, and Contextual Representations for Large-Scale Text Analytics and Topic Modelling

1. Dataset Overviews

- **20 Newsgroups:** 10,964 training documents and 7,284 testing documents across 20 distinct categories.
- **IMDb Movie Reviews:** 20,000 training and 5,000 testing documents for binary sentiment analysis.

2. Topic Modeling Results (LDA)

- **Latent Topics:** 20
- **Mean Topic Coherence Score:** 0.2040

Some representative topics discovered include:

- **Topic 1 (Discussion/Opinions):** don, think, people, just, know, like, say...
- **Topic 3 (System/OS/Software):** file, windows, software, image, graphics...
- **Topic 6 (Religion/Philosophy):** god, jesus, bible, christian, believe...
- **Topic 12 (Law/Politics):** gun, law, people, government, rights...

3. Classification & Representation Comparison

Representation	Target Dataset	Accuracy	F1-Score	Training Time
TF-IDF (Logistic Regression)	20 Newsgroups	71.06%	70.73%	18.78s
TF-IDF (Linear SVM)	20 Newsgroups	70.50%	70.28%	2.10s
Word2Vec (Logistic Regression)	20 Newsgroups	62.97%	62.37%	8.66s
GloVe (Logistic Regression)	20 Newsgroups	59.49%	58.81%	11.85s
TF-IDF (Logistic Regression)	IMDb Sentiment	88.44%	88.40%	0.65s

4. Critical Interpretation & Key Findings

1. **Lexical Dominance:** For multi-class document classification (20 Newsgroups), sparse lexical representations (**TF-IDF**) outperform averaged static embeddings (**Word2Vec / GloVe**) by around 8-11% in accuracy. This is because bag-of-words keyword presence

remains highly informative for distinct subject categories, whereas simply averaging word vectors dilutes the specific keyword signals.

2. **Sentiment Analysis:** TF-IDF exhibits highly competitive accuracy (**88.44%**) on IMDb sentiment reviews, making it a robust and lightning-fast baseline model compared to deep learning methods.